EXP.NO: 5

DATE: 1/9/26

STUDY ON ARDUINO BOARD, CONTROLLER, ANALOG & DIGITAL I/O

Aim:

To study about arduino board, Controller, Analog & digital I/O

Apparatus required:

Hardware : ARDUINO

software : ARUINO IDE

DESCRIPTION:

Digital Pins:

In addition to the specific functions listed below, the digital pins on an Arduino board can be used for general purpose input & output via the pinMode(), digitalRead() and digitalWrite() commands. Each pin has internal pull-up resistor which can be turned on and off using digital write, when the pin is configured as an input. The maximum current per pin is 40mA.

Serial: 0(RX) and 1 (TX):

used to receive (RX) and transmit (TX) TTL serial data.

External Interrupts: 2 & 3

These pins can be configured to triggger an interrupt on a low values.

PWM: 3, 5, 6, 9, 10 & 11

provide 8-Bit PWM output with analog write() function on boards with an ATMEGA 8.

BIT RESET: 7 (Arduino BT-only) to the reset line of the bluetooth module.

SPI: 10(SS), 11(MOSI), 12(MISO), 13(SCK)

These pins support SPI communication, which although provided by underlying hardware.

LED 13 :

There is a built in LED connected to digital pin 13 when the pin is High value.

Analog pins :

In addition to the specific functions listed below the analog input pins support 10-bit analog inputs can be also used as digital pins

I^2C 4(SDA) and 5(SCL)

support I^2C (IWI) communication using the were library.

power pins :

Vin : (labelled "9v"). The input voltage to the arduino board when its using an external power source (as opposed to 5V from the USB)

5V : The regulated power supply used to power microcontroller and other components.

GND : Ground pins

other pins :

AREF : Reference voltage for analog inputs not currently supported by Arunio software.

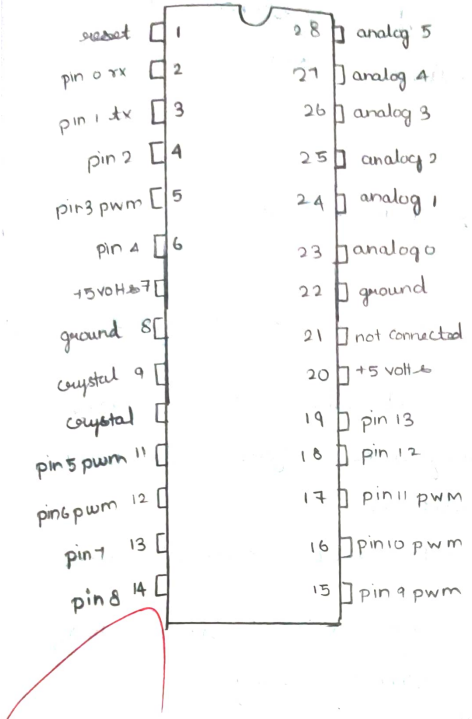
RESET : Typically to add a reset button to shields bring this line low to reset

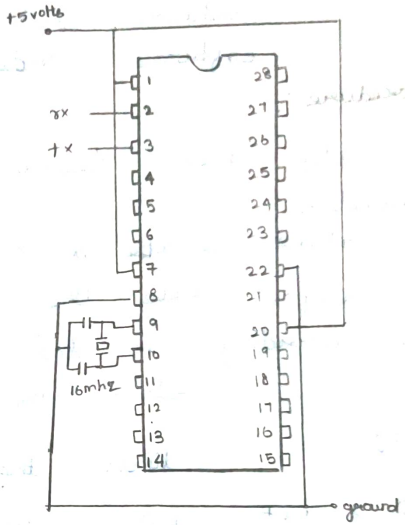
Microcontrollers

ATMEGA 328P (used on most recent boards)

Digital Ilo pins	14 (of which 6 provide PWM output)
Analog Input pins	6(DIP) or 8(SMD)
DC Current per Ilopin	40m A
Flash memory	32kB
SRAM	2kB
EEPROM	1kB

Atmega 328 microcontroller





Department of ECE		
Components	Max. Marks	Marks Obtained
Programming / Circuit Design	5	4
Execution / Circuit Connection	5	4
Results and Inferences	5	4
Viva	5	4
On time submission	5	5
Total	25	21

Result: 8/9

Thus the study of Arduino board, Controller analog and digital I/O lines is done

EXP.NO: 06 | DESIGN THE EMBEDDED SYSTEM TO INTERFAC

DATE: 8/9/26 | LED'S WITH ARDUINO UNO

AIM:

To design the embedded system to interface LEDS with arduino UNO

Apparatus required:

Hardware : Arduino

Software : Arduino IDE 2.3.5

procedure:

step1: open Arduino IDE 2.3.5 on your Computer.

step2: create a new sketch and name it

step3: write the program for Controlling LED according to the condition

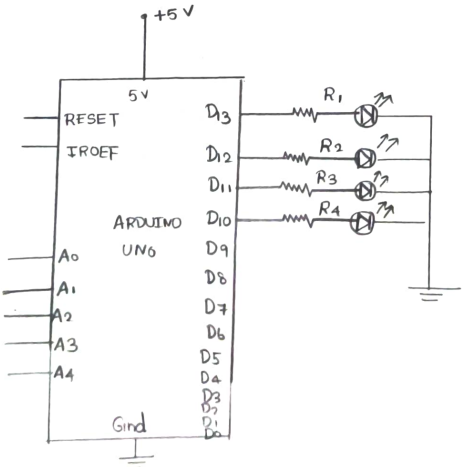
Blink the LED with 60ms time delay

Blink the LED-1 with 10ms delay LED-2 with LED-4 with 150ms delay

step4: To Compile the program select the board as arduino

step5: Connect the port & upload the program to arduino.

Circuit diagram:



a) Blink the LED'S with 60ms time delay

```
void setup()
 {
   pinmode (10, output);
   pinmode (11, output);
   pinmode (12, output);
 } pinmode (13, output);
void loop()
{
  digitalwrite (10, High);
  digitalwrite (11, High);
  digitalwrite (12, high);
  digitalwrite (13, high);
  delay (60);
  digitalwrite (10, low);
  digitalwrite (11, low);
  digitalwrite (12, low);
  digitalwrite (13, low);
  delay (60);
 }
```

b) Blink the LED1 with 10ms delay, LED-2 with 50ms delay, LED-3 with 100ms delay & LED-4 with 150ms delay

```
void setup()
{
 pinmode (10, output);
 pinmode (11, output);
 pinmode (12, output);
 pinmode (13, output);
}
void loop()
{
  digitalwrite (10, high);
  delay (1);
  digitalwrite (10, low);
  delay (1);
  digitalwrite (11, High);
 delay (50);
 digitalwrite (11, low);
 delay (50);
  digitalwrite (12, high);
```

```
    delay (100);
    digitalwrite (12, low);
    delay (100);
    digitalwrite (13, high);
    delay (150);
    digitalwrite (13, low);
    delay (150);
}
```

Department of ECE		
Components	Max. Marks	Marks Obtained
Programming / Circuit Design	5	
Execution / Circuit Connection	5	
Results and Inferences	5	
Viva	5	
On time submission	5	
Total	25	

Result:

Thus the program to design the embedded system for LED interfacing with Arduino UNO was written and verified

EXP.NO:07

DATE:13/9/26

DEVELOP THE EMBEDDED SYSTEM TO ACTIVATE Serial COMMUNICATION OF ARDUINO UNO

Aim:

To develop the embedded system to activate serial communicate of Arduino UNO.

Apparatus required:

Hardware: Arduino

Software: Arduino IDE 2.3.5

procedure:

step 1: Open Arduino IDE 2.3.5 on your Computer

step 2: Create a new sketch & name it

step 3: write the program for the folloing serial communication conditions.

a) Design the program to transmit the character "Embedded System lab" by Arduino UNO at baud rate 9600 bps

b) Design the program to transmit the character "Welcome to SONA" by Arduino at baud rate 19600 bps

c) Transmit the character 'ARDUINO LIFE' & at baud rate 4800 bps

step 4: Compile the program & select the board type as arduino

Step 5: Connect the port & upload the program.

a) Design the program to transmit the character "Embedded System Lab" by Arduino UNO at the baud rate of 9600 bps

```
void setup()
  {
    serial.begin(9600);
  }
void loop()
  {
    serial.println("Embedded System Lab");
  }
```

OUTPUT:

b) Design the program to transmit the character "Welcome to SONA" through the RS232 channel of Arduino at the baud rate of 19600 bps

```
void setup()
 {
  Serial.begin (19200);
 }
void loop()
 {
  Serial.println ("welcome to SONA");
 }
```

OUTPUT:

c) Design the program to transmit the character "Arduino Life" to the PC monitor at the rate of 4800bps

```
void setup()
 {
  Serial.begin (4800);
 }
void loop()
 {
  Serial.println ("ARDUINO LIFE");
 }
```

OUTPUT:

Department of ECE		
Components	Max. Marks	Marks Obtained
Programming / Circuit Design	5	5
Execution / Circuit Connection	5	5
Results and Inferences	5	5
Viva	5	3
On time submission	5	5
Total	25	23

Result:

Thus the experiment to develop the embedded system to activate serial communication of arduino UNO was executed successfully and result was obtained.

EXP.NO: 8

DATE: 15/9/26

DESIGN THE EMBEDDED SYSTEM TO INTERFACE KEY SWITCH, LED WITH ARDUINO UNO

Aim:

To develop the embedded system to interface Key switch, LED with arduino UNO

Apparatus Required:

Hardware : Arduino, LED-2, switch,

Software : Arduino IDE 2.3.5

procedure:

step 1: open Arduino IDE 2.3.5 on your computer

step 2: Create a new sketch & name it

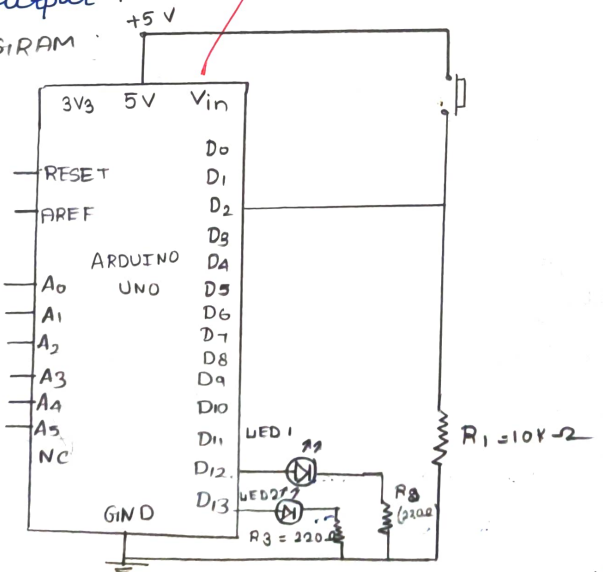
step 3: Write the program for controlling the LED using Key switch

→ Design a program that when a switch is pressed LED-1 - OFF & LED-2 → ON

step 4: Compile the program & select the board type

step 5: Connect the port & upload the program & see the output.

CIRCUIT DIAGRAM:



Design a program that when a switch is pressed LED-1 is ON and LED-2 is OFF. if switch is not pressed LED-1 is OFF & LED-2 is ON

program :

```
int buttonstate = 0 ;
void setup( )
{
    pinMode (2, input) ;
    pinMode (12, output) ;
    pinMode (13, output)
  } Serial.begin (9600) ;
void loop ( )
{
    buttonstate = digitalRead (2) ;
    if (buttonstate = = HIGH )
     {
      digitalWrite (12, HIGH ) ;
      digitalWrite (13, LOW);
     } Serial.println ("Switch is ON\n") ;
     Serial.println ("LED → 1 is ON & LED → 2 is OFF\n") ;
    else
     {
       digitalWrite (12, LOW) ;
       digitalWrite (13, HIGH) ;
     } Serial.println ("Switch is OFF\n") ;
    } Serial.println ("LED → 1 is OFF & LED → 2 is ON/n") ;
```

OUTPUT

SWITCH is ON

LED → 1 is ON & LED → 2 is OFF

SWITCH is OFF

LED → 1 is OFF & LED → 2 is ON

TWO LED - ON

```
void loop ( )
  {
    buttonstate = digitalRead (2) ;
    if (buttonstate = = HIGH)
     {
       digitalWrite (12, HIGH) ;
       digitalWrite (13, HIGH);
```

```
}
else
{ digitalWrite(12, LOW);
  digitalWrite(13, LOW);
}}
```

LED BLINK WITH POTENTIOMETER

```
int sensorpin = A0;
int ledPin = 13;
int sensorValue = 0;
void setup()
{
  pinMode(ledPin, OUTPUT);
}
void loop()
{
  int sensorValue = analogRead(sensorpin);
  digitalWrite(ledPin, HIGH);
  delay(sensorValue);
  digitalWrite(ledpin, LOW);
  delay(sensorvalue);
}
```

Arduino nano
A10 & A13 GND.

GND - GND (LED)
A12 - LED
A13 - LED
A0 → Signal
GND - GND
Vcc - 3.3V
} potentiometer

Department of ECE		
Components	Max. Marks	Marks Obtained
Programming / Circuit Design	5	
Execution / Circuit Connection	5	
Results and Inferences	5	
Viva	5	
On time submission	5	
Total	25	

Result:

Thus the program of interface key switch, LED with arduino UNO was successfully completed & the result was obtained.

EXP.NO: 09 | PWM BASED LED BRIGHTNESS AND BUZZER VOLUME CONTROL, USING ARDUINO UNO

DATE: 23/9/26

Aim:

To develop PWM based LED brightness and buzzer volume control using Arduino UNO

Apparatus required:

Hardware: Arduino, LED, potentiometer, buzzer

Software: Arduino IDE 2.3.5

procedure:

step 1: open the arduino IDE 2.3.5 software

step 2: create a new sketch & name it

step 3: write the program for the following condition

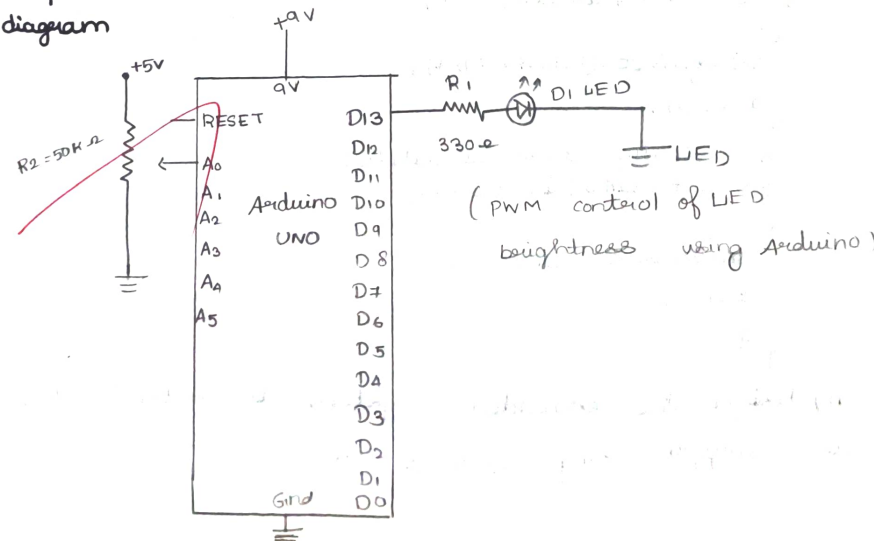
a) Design the embedded system to control the brightness of LED using potentiometer and arduino IDE.

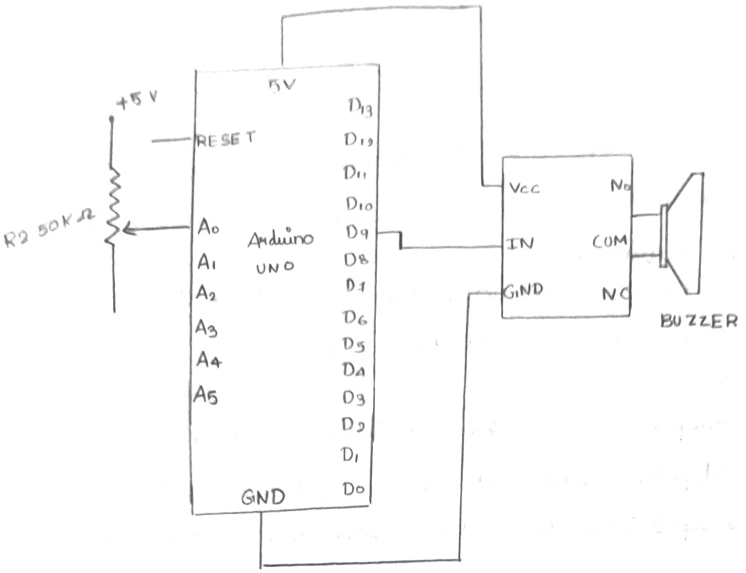
b) Design the embedded system to control the volume of buzzer using Arduino UNO

step 4: Compile the program & select the board type.

step 5: select the port & upload the program.

circuit diagram





program

a) Design the embedded system to control the brightness of LED using potentiometer & Arduino UNO

```
#define LED_PIN 11
void setup()
{
 pinMode (LED_PIN, OUTPUT);
}
void loop()
{ //Gradually increase duty cycle
 for (int i=0 ; i<255 ; i++)
  {
   analogWrite (LED_PIN , i);
   delay (5);
  }
  // Gradually decrease duty cycle
  for (int i=255 ; i>0 ; i--)
  {
   analogWrite (LED_PIN, i);
   delay (5);
  }
}
```

b) design the embedded system to control the volume of buzzer using Arduino UNO

program :

```
const int potpin = A0 ;
const int buzzerpin = 9 ;
int potvalue = 0 ;
int pwmvalue = 0 ;
void setup ()
{
    pinMode (buzzerpin, OUTPUT) ;
    Serial.begin (9600) ;
}
void loop ()
{
    potvalue = analogRead (potpin) ;
    pwmvalue = map (potvalue, 0, 1023, 0, 255) ;
    analogWrite (buzzerPin, pwmvalue) ;
    Serial.println ("Potentiometer value:") ;
    Serial.print (potvalue) ;
    Serial.println ("PWM value :") ;
    Serial.print (pot value) ;
    Serial.println ("BUZZER : ON CONDITION") ;
}
```

delay (100);

```
const int buzzer = 13 ;
void setup() {
  pinMode (buzzer, OUTPUT);
}
void loop() {
  digitalWrite (buzzer, HIGH);
  delay (500) ;
  digitalWrite (buzzer, LOW);
  delay (500);
}
```

Department of ECE		
Components	Max. Marks	Marks Obtained
Programming / Circuit Design	5	
Execution / Circuit Connection	5	
Results and Inferences	5	
Viva	5	
On time submission	5	
Total	25	

Result :

Thus the program PWM based LED brightness and buzzer volume control using arduino was successfully completed and result was obtained.